

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 Total Marks: 25

Annexure A **APPLICATION BASED QUESTIONS**

Code of Conduct:

I X. Joshua (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

Short Answer Test

Q1. Explain all the possible blocks in a Machine Learning System. Give examples of Supervised and Unsupervised Learning.

Answer

A **Machine Learning (ML) system** learns patterns from data and makes predictions or decisions without being explicitly programmed.

Block Diagram

Data Collection



Data Preprocessing



Feature Selection / Feature Engineering



Model Selection



Model Training



Model Evaluation



Prediction / Deployment

Explanation of Each Block

1. Data Collection

- Data is collected from databases, sensors, websites, or files.
- Example: Customer purchase records.

2. Data Preprocessing

- Removes missing values, duplicate records, and noise.
- Converts raw data into a clean format.

3. Feature Selection

- Selects only important features.
- Reduces complexity and improves accuracy.
- Example: Selecting Age and Income while removing unnecessary attributes.

4. Model Selection

- Choose a suitable algorithm.
- Examples:

- Decision Tree
- Naïve Bayes
- SVM
- Random Forest

5. Model Training

- The algorithm learns from training data.
- Model parameters are adjusted to reduce prediction errors.

6. Model Evaluation

- Performance is tested using unseen data.
- Metrics:
- Accuracy
- Precision
- Recall
- F1-score

7. Prediction/Deployment

- The trained model is used for real-world predictions.
- Example: Predicting loan approval.

Supervised Learning

- Uses **labelled data**.
- Learns input-output relationship.

Example:

Input	Output
Student Marks	Pass/Fail

Algorithms:

- Decision Tree
- Logistic Regression
- Naïve Bayes

Unsupervised Learning

- Uses **unlabelled data**.
- Finds hidden patterns.

Example:

- Customer segmentation using K-Means.

Algorithms:

- K-Means
- Hierarchical Clustering

Conclusion

Machine Learning consists of data collection, preprocessing, feature selection, model training, evaluation and deployment. Supervised learning predicts known outputs, whereas unsupervised learning discovers hidden patterns.

Q2. Design a Machine Learning System for Disaster Management

Answer

A disaster prediction system helps predict floods and earthquakes using historical and real-time sensor data.

System Architecture

Historical Data + Real-Time Sensors



Data Preprocessing



Feature Selection



Machine Learning Model



Prediction Result



Early Warning & Emergency Response

Model Choice

Random Forest is suitable because:

- High prediction accuracy
- Handles missing values
- Works well with large datasets
- Resistant to overfitting

Data Preprocessing

- Remove missing values

- ☐ Remove duplicate records
- ☐ Normalize data
- ☐ Convert categorical values into numerical values

Feature Selection

Important features:

- ☐ Rainfall
- ☐ River Water Level
- ☐ Soil Moisture
- ☐ Temperature
- ☐ Seismic Activity
- ☐ Ground Vibration

(a) Training, Validation and Evaluation

Training

- ☐ Train model using historical disaster data.

Validation

- ☐ Tune model parameters.

Evaluation

- ☐ Test using unseen data.

Metrics:

- ☐ Accuracy
- ☐ Precision
- ☐ Recall
- ☐ F1-score

(b) Early Warning and Emergency Response

The system can:

- ☐ Predict floods before occurrence.
- ☐ Detect earthquake risks.
- ☐ Send SMS alerts.
- ☐ Notify government agencies.
- ☐ Help evacuation planning.
- ☐ Reduce loss of life and property.

Conclusion

Machine learning improves disaster prediction and enables faster emergency response using real-time monitoring.

Q3. Apply Find-S Algorithm (Play Tennis Dataset)

Positive examples only:

Example	Outlook	Temperature	Humidity	Wind
3	Overcast	Hot	High	Weak
4	Rainy	Mild	High	Weak
5	Rainy	Cool	Normal	Weak
7	Overcast	Mild	Normal	Strong

Step 1

Initial Hypothesis

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Step 2 (Example 3)

$S_1 = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$

Step 3 (Example 4)

Compare:

Outlook:

Overcast $\neq$ Rainy $\rightarrow ?$

Temperature:

Hot $\neq$ Mild $\rightarrow ?$

Humidity:

High = High

Wind:

Weak = Weak

$S_2 = \langle ?, ?, \text{High}, \text{Weak} \rangle$

Step 4 (Example 5)

Humidity:

High $\neq$ Normal $\rightarrow$?

Wind:

Weak = Weak

S3 = <?, ?, ?, Weak>

Step 5 (Example 7)

Wind:

Weak $\neq$ Strong $\rightarrow$?

S4 = <?, ?, ?, ?>

Final Hypothesis

<?, ?, ?, ?>

Meaning:

Every attribute becomes generalized because the positive examples differ in every attribute.

Q4. Candidate Elimination Algorithm

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Initialization

S0=<∅,∅,∅,∅>

G0=<?,?,?,?>

Example 1 (Positive)

S=<Sunny,Warm,Normal,Weak>

G=<?,?,?,?>

Example 2 (Positive)

Humidity differs

Wind differs

S=<Sunny,Warm,?,?>

Example 3 (Negative)

Specialize G to exclude negative example.

G=<Sunny,?,?,?>

Example 4 (Positive)

Temperature differs

S=<Sunny,?, ?,?>

Example 5 (Negative)

Already excluded by Weather.

No further change.

Final Boundaries

Specific Boundary

S=<Sunny,?,?,?>

General Boundary

G=<Sunny,?,?,?>

Version Space

Contains all hypotheses between S and G.

Q5. Find-S Algorithm (Customer Purchase Dataset)

Positive examples:

Example	Age	Income	Student	Credit Rating
3	Middle	High	No	Fair
4	Senior	Medium	No	Fair

Initial Hypothesis

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Example 3

$S_1 = \langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$

Example 4

Age differs

Income differs

Student same

Credit Rating same

$S_2 = \langle ?, ?, \text{No}, \text{Fair} \rangle$

Final Hypothesis

$\langle ?, ?, \text{No}, \text{Fair} \rangle$

Interpretation

The customer is likely to buy a product if:

- Student = No
- Credit Rating = Fair
- Age and Income can be any value

Q6. Apply the Find-S Algorithm (Workout Dataset)

Dataset

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

Only **positive examples** (2, 3, and 4) are considered.

Step 1: Initialize

Initial hypothesis:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Step 2: Process Example 2 (Positive)

$S_1 = \langle \text{Rainy, Mild, Evening, Happy} \rangle$

Step 3: Process Example 3 (Positive)

Compare with Example 3:

- ☐ Weather: Rainy $\neq$ Sunny $\rightarrow ?$
- ☐ Temperature: Mild = Mild
- ☐ Time: Evening = Evening
- ☐ Mood: Happy = Happy

Updated hypothesis:

$S_2 = \langle ?, \text{Mild, Evening, Happy} \rangle$

Step 4: Process Example 4 (Positive)

Compare with Example 4:

- ☐ Weather: ? remains ?
- ☐ Temperature: Mild $\neq$ Cool $\rightarrow ?$
- ☐ Time: Evening $\neq$ Morning $\rightarrow ?$
- ☐ Mood: Happy = Happy

Updated hypothesis:

$S_3 = \langle ?, ?, ?, \text{Happy} \rangle$

Final Hypothesis

<?, ?, ?, Happy>

Conclusion

The Find-S algorithm concludes that the only common condition among all positive examples is that the person's **Mood = Happy**. Weather, Temperature, and Time can be any value.

Q7. Apply the Find-S Algorithm (Loan Approval Dataset)

Dataset

| Example | Employment | Salary | Credit Score | Loan History | Loan Approved |
|---------|------------|--------|--------------|--------------|---------------|
| 1 | Employed | High | Good | Clean | Yes |
| 2 | Employed | Medium | Good | Clean | Yes |
| 3 | Unemployed | Low | Poor | Bad | No |
| 4 | Employed | Medium | Excellent | Clean | Yes |

Only positive examples (1, 2, and 4) are used.

Step 1: Initialize

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Step 2: Example 1

$S_1 = \langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$

Step 3: Example 2

- Employment: Same
- Salary: High $\neq$ Medium $\rightarrow$?
- Credit Score: Same
- Loan History: Same

$S_2 = \langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$

Step 4: Example 4

- Employment: Same
- Salary: ? remains ?
- Credit Score: Good $\neq$ Excellent $\rightarrow$?
- Loan History: Same

$S_3 = \langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Final Hypothesis

$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Conclusion

A loan is likely to be approved if the applicant is **Employed** and has a **Clean Loan History**, while Salary and Credit Score may vary.

Q8. Candidate Elimination Algorithm (Customer Subscription)

Initialization

Specific Boundary:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

General Boundary:

$G_0 = \langle ?, ?, ?, ? \rangle$

Example 1 (Positive)

$S = \langle \text{Young}, \text{High}, \text{Frequent}, \text{Mobile} \rangle$

$G = \langle ?, ?, ?, ? \rangle$

Example 2 (Positive)

Only Internet Usage differs.

$S = \langle \text{Young}, \text{High}, ?, \text{Mobile} \rangle$

Example 3 (Negative)

Negative example:

$\langle \text{Middle}, \text{Low}, \text{Frequent}, \text{Desktop} \rangle$

Specialize G to exclude this example.

Possible G:

$\langle \text{Young}, ?, ?, ? \rangle$

Example 4 (Positive)

Income differs.

$S = \langle \text{Young}, ?, ?, \text{Mobile} \rangle$

Example 5 (Negative)

Already excluded because Age Group = Senior.

No further change.

Final Boundaries

Specific Boundary (S)

<Young, ?, ?, Mobile>

General Boundary (G)

<Young, ?, ?, ?>

Version Space

The version space contains all hypotheses between S and G that correctly classify the training data.

Q9. Candidate Elimination Algorithm (Healthy Lifestyle)

Initialization

Specific Boundary:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

General Boundary:

$G_0 = \langle ?, ?, ?, ? \rangle$

Example 1 (Positive)

$S = \langle \text{Balanced, Regular, Adequate, Low} \rangle$

Example 2 (Positive)

Sleep differs.

Stress differs.

$S = \langle \text{Balanced, Regular, ?, ?} \rangle$

Example 3 (Negative)

Negative example:

<Junk, None, Poor, High>

Specialize G to exclude the negative example.

$G = \langle \text{Balanced, ?, ?, ?} \rangle$

Example 4 (Positive)

Exercise differs.

S = <Balanced, ?, ?, ?>

Example 5 (Negative)

Already rejected because Diet = Junk.

No change.

Final Boundaries

Specific Boundary:

<Balanced, ?, ?, ?>

General Boundary:

<Balanced, ?, ?, ?>

Conclusion

The algorithm concludes that a **Balanced Diet** is the most important attribute for predicting a healthy lifestyle, while the remaining attributes may vary.

Q10. Design a Machine Learning System for Health Analysis

Answer

A Machine Learning health analysis system predicts diseases such as **heart disease** or **diabetes** using patient information.

System Architecture



Suitable Algorithms

1. Logistic Regression

- Suitable for disease classification.
- Fast and easy to interpret.

2. Decision Tree

- Produces simple decision rules.
- Easy for doctors to understand.

3. Random Forest

- High prediction accuracy.
- Reduces overfitting.
- Handles large datasets effectively.

Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize numerical data.
- Encode categorical variables.
- Split data into training and testing sets.

Feature Selection

Important features include:

- Age
- Blood Pressure
- Blood Sugar (Glucose)
- Cholesterol
- BMI
- Smoking Habit
- Physical Activity
- Family History

Feature selection reduces unnecessary attributes, improves accuracy, and decreases training time.

Model Training

1. Split the dataset into training and testing sets.
2. Train the selected algorithm.

3. Validate the model using validation data.
4. Evaluate performance using test data.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Advantages

- Early disease detection.
- Better treatment planning.
- Reduces diagnostic errors.
- Supports doctors in clinical decision-making.
- Improves patient care.

Conclusion

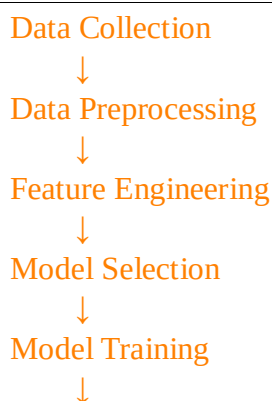
A health analysis system using **Random Forest or Logistic Regression**, combined with proper preprocessing and feature selection, provides accurate disease prediction and supports timely medical intervention

Q11. Explain the Architecture of a Typical Machine Learning Pipeline. Differentiate between Training Error and Generalization Error.

Answer

A **Machine Learning Pipeline** is a sequence of steps used to build, train, evaluate, and deploy a machine learning model. It ensures that data flows systematically from collection to prediction.

Architecture of a Machine Learning Pipeline



Model Evaluation



Model Deployment



Prediction

Stages of the Pipeline

1. Data Collection

- Data is collected from sensors, databases, websites, or IoT devices.
- Example: Collecting patient records for disease prediction.

2. Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize numerical data.
- Encode categorical attributes.

3. Feature Engineering

- Select important features.
- Create new useful features.
- Reduce dimensionality using PCA if required.

4. Model Training

- Train the selected algorithm using training data.
- Common algorithms:
 - Decision Tree
 - Random Forest
 - Naïve Bayes
 - SVM

5. Model Evaluation

Evaluate the model using:

- Accuracy
- Precision
- Recall
- F1-Score

6. Model Deployment

- Deploy the trained model to a website, mobile app, or cloud server.
- The model predicts outcomes for new data.

Difference between Training Error and Generalization Error

Training Error	Generalization Error
Error on training data	Error on unseen test data
Usually low	Measures real-world performance
Does not indicate actual performance	Indicates model's ability to predict new data
Example: 98% accuracy on training data	Example: 90% accuracy on test data

Example

A spam detection model achieves:

- Training Accuracy = **99%**
- Testing Accuracy = **91%**

The difference indicates **generalization error**.

Conclusion

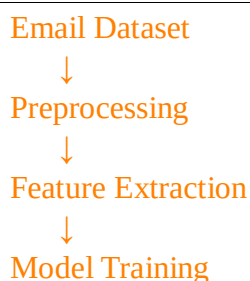
A good machine learning pipeline produces reliable models. Low training error and low generalization error indicate good performance.

Q12. Design a Machine Learning Model for Spam Email Classification

Answer

Spam email classification identifies whether an email is **Spam** or **Not Spam** using labelled data.

System Architecture



↓
Prediction
↓
Spam / Not Spam

(a) Choice of Algorithm

Naïve Bayes

Reasons:

- Fast training.
- Works well with text data.
- High accuracy.
- Uses probability for classification.

Other suitable algorithms:

- Support Vector Machine (SVM)
- Logistic Regression

(b) Preprocessing Steps

1. Tokenization

Split the email into individual words.

Example:

"Win a free phone"
↓
Win | a | free | phone

2. Stop-word Removal

Remove common words such as:

- is
- the
- a
- an
- of

This reduces unnecessary information.

3. Stemming/Lemmatization

$$F1 = \frac{2PR}{P+R}$$

Convert words to their root form.

Example:

- Playing → Play
- Running → Run

4. Vectorization

Convert text into numerical values using:

- Bag of Words
- TF-IDF

This allows machine learning algorithms to process text.

(c) Evaluation Metrics

Accuracy

Percentage of correctly classified emails.

Precision

Measures how many emails predicted as spam are actually spam.

Recall

Measures how many actual spam emails are detected.

F1-Score

Harmonic mean of Precision and Recall.

$$F1 = \frac{2PR}{P+R}$$

Advantages

- Filters unwanted emails.
- Saves user time.
- Improves email security.
- Reduces phishing attacks.

Conclusion

Naïve Bayes with proper preprocessing provides an efficient and accurate spam email classification system.

Q13. Candidate Elimination Algorithm (EnjoySport Dataset)

(a) Initialization of S and G

The Candidate Elimination Algorithm maintains two boundaries:

Specific Boundary (S)

Most specific hypothesis.

$S = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$

General Boundary (G)

Most general hypothesis.

$G = \langle ?, ?, ?, ?, ?, ? \rangle$

(b) Updating S and G

Step 1

For the first positive example:

Update S to match the example.

Example:

S =
<Sunny,
Warm,
Normal,
Strong,
Warm,
Same>

G remains unchanged.

Step 2

For another positive example:

Generalize S wherever attribute values differ.

Example:

S =
<Sunny,
Warm,
?,
Strong,
?,
Same>

Step 3

For a negative example:

Specialize G to exclude the negative example while remaining consistent with S.

Example:

G =
<Sunny,
?,
?,
?,
?,
?>

Final Version Space

The version space consists of all hypotheses between the final **Specific Boundary (S)** and **General Boundary (G)**.

(c) Comparison between Candidate Elimination and Find-S

Find-S	Candidate Elimination
Uses only positive examples	Uses both positive and negative examples
Finds one maximally specific hypothesis	Finds all consistent hypotheses
Simpler algorithm	More comprehensive algorithm
Does not maintain version space	Maintains version space (S and G)

Find-S

Candidate Elimination

Less accurate for complex datasets	Better learning capability
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Conclusion

Candidate Elimination is more powerful because it maintains both the most specific and most general hypotheses.

Q14. Machine Learning System for House Price Prediction

Answer

House price prediction estimates the selling price of a house using features such as location, size, number of rooms, and amenities.

System Architecture

House Data
(Location, Size, Rooms)



Preprocessing



Feature Selection



Regression Model



Price Prediction

(a) Choice of Regression Model

Linear Regression

Suitable because:

- Simple to implement.
- Predicts continuous values.
- Easy to interpret.

Decision Tree Regression

- Handles nonlinear relationships.
- Easy to visualize.

Random Forest Regression

- High prediction accuracy.
- Handles complex datasets.
- Reduces overfitting.

Random Forest is generally preferred because it provides better performance.

(b) Feature Scaling and Missing Data

Feature Scaling

Scaling converts features into the same range.

Methods:

- Min-Max Normalization
- Standardization

Benefits:

- Faster training.
- Better model performance.

Handling Missing Data

Methods:

- Remove incomplete records.
- Replace missing values using:
 - Mean
 - Median
 - Mode

(c) Detecting and Avoiding Overfitting

Detecting Overfitting

Symptoms:

- Very high training accuracy.
- Low testing accuracy.

Avoiding Overfitting

- Cross-validation.
- Feature selection.

- Increase training data.
- Regularization.
- Random Forest.
- Pruning (Decision Trees).

Conclusion

Random Forest Regression combined with proper preprocessing, feature scaling, and overfitting control provides accurate house price prediction.

Q15. Explain Clustering in Unsupervised Learning

Answer

Clustering is an **unsupervised learning** technique that groups similar data points into clusters without using labelled data.

Applications:

- Customer segmentation
- Image segmentation
- Document clustering
- Market analysis

(a) Working of K-Means Algorithm

Steps

Step 1: Choose the number of clusters (**K**).

Step 2: Select K initial centroids randomly.

Step 3: Assign each data point to the nearest centroid.

Step 4: Calculate the new centroid of each cluster.

Step 5: Repeat Steps 3 and 4 until the centroids no longer change.

Example

Suppose we have student marks:

20, 25, 30, 80, 85, 90

Choose **K = 2**

Initial centroids:

- $C_1 = 25$

$$\square \quad C_2 = 85$$

Cluster Assignment:

Cluster 1:

20, 25, 30

Cluster 2:

80, 85, 90

New Centroids:

$$\square \quad C_1 = 25$$

$$\square \quad C_2 = 85$$

Since the centroids do not change further, the algorithm stops.

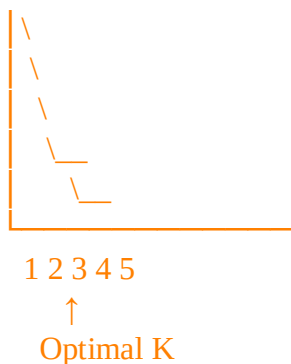
(b) Choosing K using the Elbow Method

The Elbow Method helps determine the optimal value of **K**.

Steps:

1. Run K-Means for different values of K.
2. Calculate the Within-Cluster Sum of Squares (WCSS).
3. Plot **K** versus **WCSS**.
4. Choose the point where the graph forms an "elbow."

WCSS



The elbow point indicates the best number of clusters.

(c) K-Means vs Hierarchical Clustering

K-Means	Hierarchical Clustering
Requires K before clustering	Does not require K initially
Faster for large datasets	Slower for large datasets

K-Means	Hierarchical Clustering
Uses centroids	Uses dendrogram (tree structure)
Suitable for large datasets	Suitable for small datasets
Produces flat clusters	Produces hierarchical clusters

Conclusion

Clustering is a fundamental unsupervised learning technique. K-Means is efficient for large datasets, while Hierarchical Clustering provides a tree-like representation and is useful for smaller datasets